Rank-27 Congruence Membership-Range Repair

Tactix Development Project

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Executive summary

This increment promotes a general representation repair in Congruence's immutable small-clause index. Canonical records are already lexicographically sorted, so a queried key's first literal identifies one contiguous interval. The new directory restricts the existing binary search to that interval. It does not alter record or gate order, candidate admission, logical ticks or costs, scheduling, proof transactions, learning, pruning, or N1–N4.

The exact conflict-9450 Rank-27 profile reduces complete sampled cycles from 64.668B to 63.456B and the intended extraction owner from 2.914B to 1.355B. All 64 event-trace rows are byte-identical. The complete frozen B28 plus fixed C5 guard passes 66/66 independently checked tasks, and corrected exact-source ASan/UBSan native-LRAT fuzz passes 10,000 cases with zero failures. Checked Rank-27 job `1645875` verifies the byte-identical 1,514,395,111-byte DRAT proof and removes it, improving internal time 407.006 to 403.855 seconds. The repair is accepted, but Rank 27 remains open because the frozen target is 300 seconds.

Frozen Kissat 4.0.4 and all other frozen reference solvers were not rerun. The user-canceled full Tier-D arrays remain frozen, non-scorable, and residue-free; this increment uses only the predeclared Rank-27 input and mandatory promotion guards.

1 Authority and localization

The immutable input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. Every Rank-27 job uses two CPUs and the 16 GiB/1200 s contract. The accepted predecessor is checked job `1645778`: 407.006 seconds internal, 407.497 seconds wall, and proof SHA-256 `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d`. Current instrumented Kissat is source-observer evidence only; its accepted diagnostic run is not a scoring baseline.

An initial Kissat-shaped complementary-condition pair index was not retained. Paired stage control `1645798` preserved the state but made ITE extraction 928 to 974 ms, complete Congruence 2739 to 2783 ms, and the whole prefix 16.940 to 17.041 seconds. This rejects pair-list sorting absent contradictory population evidence.

Temporary observer `1645800` then divided ITE extraction without changing behavior. Of 849 cumulative milliseconds, conditional-equivalence collection owned 635 ms (74.8%), canonical pair preparation 68 ms, and admitted-pair evaluation 76 ms. Source tracing found that collection performs repeated full-index membership searches for canonical XOR/ITE clause keys. Because each key is sorted, its first literal is a portable pre-existing partition of the sorted record vector.

2 Repair and exact common-state evidence

After index construction and deduplication, the candidate builds one `uint32_t` boundary per internal literal. `contains` applies the unchanged comparator and `lower_bound` only between the two boundaries for the key's first literal. The directory is scoped to its immutable index; it is neither a live-clause authority nor a policy input. Frozen Release pin SHA-256 is `b81bb832b81350e5269fd9ff750013c929320e8a9c141a00cd7c04e581c51b83`, from base `fff43d69` and source-diff SHA-256 `283d6cce79848e230ccd2ddf5a1647e0a5b4c18702d6c155522fbf4c0af1eb9d`.

Evidence	Predecessor	Candidate	Result
ITE extraction, ms	1007	550	0.54618x
XOR extraction, ms	168	91	0.54167x
Complete Congruence, ms	2853	2259	0.79180x
Conflict-9450 solve, ms	16869	16512	0.97884x
Complete sampled cycles	64.668B	63.456B	0.98125x
Extraction-owner cycles	2.914B	1.355B	0.46494x

Stage job `1645801` and cycles job `1645803` run paired arms at the same conflict-9450 state. Every printed non-time statistic is exact, including 9,450 conflicts, 5,795,248 decisions, 96,338,812 propagations, 1,734,747 gates, 58,738 merges, 181 units, 445,868,596 Congruence ticks, and 1,744,090 closure items. Both cycle profiles report zero lost samples. The eight-millisecond index-build increase is dominated by the 594-millisecond complete-Congruence reduction.

Trace job `1645805` records 64 untruncated rows from both arms. The files are byte-identical with SHA-256 `7a6216bb86c236512dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1`. Thus rule-boundary digest, candidate identity and order, exact tick/cost image, scheduling order, learning, pruning, and resulting state remain common through the conflict-9449 Probe and conflict-9450 stop.

3 Correctness and regression gates

Release validation passes 73 of 74 tests. The only failure is the previously recorded opt-in documentation-size debt; all behavioral, proof/model, cleanup, trace, API, N1, and rule/policy-boundary tests pass. Focused Debug ASan/UBSan Kernel, Congruence, event-trace, random-policy, micro-proof, solve-reset, and rule/policy-boundary tests pass with leak detection disabled because the execution wrapper prevents LeakSanitizer startup.

Full-width campaign `benchmarks/runs/20260824-1685-BC-first-literal-range-guard/`, array `1645806` and aggregate `1645872`, validates both immutable arms on all 28 Tier-B and five predeclared mechanism-relevant Tier-C instances. Each arm passes 17 SAT models and 16 UNSAT proofs; there is no solve loss, invalidity, regression, cleanup failure, or certificate residue. After timing lines are removed, all 33 paired solver-statistic files are byte-identical. Candidate PAR-2 is 1700.709 versus 1705.542 seconds, ratio 0.9971663. Aggregate summary SHA-256 is `e94a30b5e30e009b12539f370cd3c761fa370689eb88ce0fb6f53eee258025cc`.

The first targeted fuzz submission, `1645873`, is frozen setup-invalid: Congruence-only stress starved one Sweep-specific corpus proof case, and the harness correctly refused to begin random fuzzing. It had zero corpus failures and no solver correctness finding. Corrected job `1645874` pre-registers exactly that corpus-only nontermination and retains strict random validation. It completes in 01:11:16 with 10,000 cases and zero failures: 2,620 checked SAT models, 3,482 checked native-LRAT UNSAT proofs, and 3,898 bounded five-second nonterminations on which no oracle claim is made. Congruence fires in 3,556 cases for 1,246,344 target applications. There is no sanitizer

finding, rejected model/proof, crash, certificate residue, or job temporary residue. The 25 retained timeout CNFs are declared diagnostic inputs, not certificates. Log SHA-256 is `8d33c4a4a43818158 7d8d5d466bd7c2f19bf201fdeab4fe10484642b47a3db16`.

4 Checked Rank-27 result and disposition

Immutable campaign `benchmarks/runs/20260824-1695-tierd-rank27-first-literal-range -witness/`, job `1645875`, completes on two CPUs in 00:08:40. Tactix reports UNSAT in 403.855 internal / 404.395 wall seconds with 613,652 KiB peak solver RSS. Pinned `drat-trim` reads the original CNF and the 1,514,395,111-byte proof, reports `s VERIFIED`, and exits zero after 112.308 internal / 112.489 wall seconds. The proof SHA is byte-identical to the predecessor. Proof, CNF, checker derivatives, and scratch are deleted before the atomic valid result; all cleanup flags are true and residue is zero.

Removing timing lines makes every solver statistic byte-identical to `1645778`, normalized SHA-256 `8cd3eda481799871a179e1483a57413622bc7832c09432df705b2308bafc2da7`. Internal/wall improves 407.006/407.497 to 403.855/404.395 seconds, a recovery of 3.151/3.102 seconds on the exact trajectory. Congruence timing falls 18.130 to 16.910 seconds. The repair therefore passes promotion; it does not close Rank 27, whose checked result remains 1.346x the 300-second target.

Next work returns to current Tactix versus current instrumented Kissat and the earliest common-state event where Tactix performs slower comparable work or retains weaker state. The broad search/Vivify throughput lead remains open. All prior Sweep semantic/throughput, counted directed-bridge, BVE antecedent- admission, proof-rule/strategy, reserved-trail, rejected propagation, and N1–N4 closures remain in force. Frozen references and canceled full Tier-D campaigns remain untouched.